# A Reliability-Oriented and Interpretable Machine Learning Framework for Diabetes Analytics Using the CDC BRFSS Dataset

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**Abstract**

Machine-learning models based on self-reported health indicators can support diabetes risk screening, but their evaluation requires more than reporting classification accuracy because class imbalance, decision-threshold choice, probability calibration, and model interpretation can lead to different conclusions about model performance. This study analyzed 253,680 records from the unweighted CDC BRFSS 2015 Diabetes Health Indicators dataset, in which reported prediabetes or diabetes accounted for 13.93% of the sample. Logistic Regression, Decision Tree, Random Forest, and XGBoost were compared using five-fold stratified cross-validation on an 80% development set, with PR-AUC used as the primary model-selection metric. XGBoost achieved the highest mean cross-validated PR-AUC of 0.4359 and was selected for subsequent evaluation. To adapt the classifier to a screening setting, the decision threshold was selected exclusively from out-of-fold development predictions under a prespecified Recall $\geq$ **0.80** constraint, resulting in a threshold of 0.13 before the independent internal holdout set was examined. On the holdout set, the final model achieved a PR-AUC of 0.4238 and ROC-AUC of 0.8272; at the selected threshold, Recall reached 80.99%, Precision 29.91%, and F1-score 0.4369. The predicted probabilities had a Brier score of 0.0974, a calibration slope of 0.9590, and an intercept of $-$**0.0514**. SHAP ranked self-rated general health, high blood pressure, age, BMI, and high cholesterol as the five largest contributors to model predictions. Comparison with univariate statistical effect-size rankings showed seven shared features in the Top-10, with a Jaccard similarity of 0.5385 and a Spearman correlation of 0.7098, indicating partial rather than complete agreement between marginal association and multivariable predictive contribution. Taken together, the analysis separates model selection, threshold choice, holdout evaluation, calibration, and interpretation within a single development–evaluation sequence while avoiding the use of holdout performance to guide development decisions. The findings apply to the analyzed BRFSS sample and should not be interpreted as clinical diagnosis, causal evidence, or population-level estimates.

**Keywords:** Diabetes Risk Screening, BRFSS, Machine Learning, XGBoost, Calibration, SHAP, Statistical Association, Explainable Artificial Intelligence

# 1 Introduction

Diabetes is associated with cardiovascular, renal, neurological, and vision-related complications, making early identification of individuals at elevated risk relevant to screening and preventive care [16, 22, 24]. The Behavioral Risk Factor Surveillance System (BRFSS) provides self-reported demographic, behavioral, and health indicators that can be used to study diabetes risk without laboratory measurements. Machine-learning methods have therefore been applied to BRFSS and related datasets using variables such as age, body mass index, blood pressure, cholesterol status, physical activity, and self-rated health [7, 15, 23].

Model performance in this setting cannot be judged from Accuracy alone. In the BRFSS 2015 dataset analyzed here, the positive class accounts for only **13.93%** of the records. A classifier can therefore achieve high Accuracy while detecting only a small proportion of positive cases. Metrics such as PR-AUC and Recall are more informative when positive-case detection is a central objective [19]. The classification threshold adds another decision point because increasing Recall generally changes Precision, Specificity, and the number of false-positive predictions. For this reason, threshold selection should be separated from final holdout evaluation rather than adjusted after observing test performance.

Discrimination alone also does not determine whether predicted probabilities are reliable. A model may rank positive cases above negative cases while still producing probabilities that do not correspond closely to observed outcome frequencies. Calibration therefore needs to be evaluated separately using measures such as the Brier score, calibration slope, and calibration intercept [6].

Interpretability introduces a related but distinct issue. SHAP can quantify how predictors contribute to the output of a fitted model [9, 10], but a large SHAP value does not imply a strong marginal statistical association or a causal effect. Predictors may share information, interact with other variables, or become more or less useful after other features are considered. Comparing model-based importance with statistical effect sizes can therefore show where these two forms of evidence agree and where they differ.

Based on these considerations, this study evaluates a single analysis pipeline on the original imbalanced CDC BRFSS 2015 Diabetes Health Indicators dataset. Four classifiers are compared using five-fold stratified cross-validation on a development set, with PR-AUC used for model selection. The selected model is then assigned a screening threshold using only out-of-fold development predictions under a prespecified Recall constraint. After the model and threshold are fixed, performance is evaluated on an independent internal holdout set using discrimination, classification, bootstrap uncertainty, and calibration measures. The final model is also examined with SHAP, and its global feature ranking is compared with univariate statistical effect-size rankings.

The purpose of this workflow is not to introduce a new classifier or to treat SHAP as clinical evidence. Instead, the study examines whether model selection, threshold choice, probability calibration, uncertainty estimation, and statistical interpretation can be kept within one evaluation sequence without using the holdout set to guide development decisions. The reported findings are limited to the analyzed, unweighted BRFSS sample and should not be interpreted as clinical diagnosis, causal evidence, or population-level estimates. All data-processing scripts, SQL validation queries, statistical analyses, the machine-learning pipeline, calibration and SHAP analyses, figures, and reproducibility checks are publicly available at https://github.com/AI-Peak/Diabetes-Analytics.

# 2 Related Work

Machine learning has been widely applied to diabetes risk prediction using survey, clinical, and laboratory data. Using BRFSS 2014, Xie et al. [23] compared several

classifiers, including Logistic Regression, Decision Tree, Random Forest, Neural Network, and Support Vector Machine, and also examined adjusted associations using multivariable regression. More recent BRFSS-based studies have evaluated ensemble methods and explainability techniques. Li et al. [7] combined class-balancing methods, GA-optimized XGBoost, stacking, and SHAP, while other studies have used tree-based models and SHAP to examine diabetes-related predictors in BRFSS data [11, 15]. These studies show that both predictive modeling and model explanation are already established components of BRFSS-based diabetes research; broader reviews also emphasize substantial heterogeneity in datasets, preprocessing, validation, and reporting practices [13].

Evaluation choices remain important because diabetes datasets are commonly imbalanced and different metrics describe different aspects of model performance. Precision–Recall analysis is more informative than ROC analysis alone when the positive class is relatively uncommon [19]. Prediction-model guidance also distinguishes model development decisions from final evaluation and recommends assessing performance beyond a single discrimination metric [3, 5]. Reporting and risk-of-bias guidance for prediction models further stresses transparent separation of model development, evaluation, and applicability claims [4, 14]. Calibration is a separate property from discrimination and should be examined when predicted probabilities are interpreted as risk estimates [6].

Interpretability introduces a second distinction. SHAP can quantify how individual features contribute to predictions from tree-based models [9, 10], and related healthcare studies have used SHAP to interpret diabetes and glycemic prediction models [8, 18]. However, SHAP importance is not equivalent to statistical association or causal effect. A feature may have a strong marginal association with the outcome while adding limited information after other predictors are considered, and a feature with modest marginal association may still contribute to a multivariable model.

Accordingly, the present study does not treat the use of XGBoost or SHAP as a methodological novelty. Its focus is the evaluation sequence applied to the original imbalanced BRFSS 2015 sample: model selection by cross-validation, threshold selection from out-of-fold development predictions, one-time internal holdout evaluation, bootstrap uncertainty estimation, calibration assessment, and comparison between SHAP importance and statistical effect-size rankings. The statistical–SHAP comparison is used as an exploratory analysis of agreement between marginal association and predictive contribution, not as a validation of causal or clinical importance.

# 3 Materials and Methods

Figure 1 summarizes the analysis pipeline. The workflow comprised data validation, statistical association analysis, model development and selection, threshold selection, holdout evaluation, calibration assessment, and SHAP-based explanation. The complete workflow can be executed end to end through the repository-level pipeline script, and the figures and tables reported below are regenerated by the same code.

## 3.1 Dataset and Data Validation

The CDC BRFSS 2015 Diabetes Health Indicators dataset contained **253,680 records** and 21 predictors [1, 20, 21]. The original data sources are documented in the project repository. The binary target combined reported prediabetes and diabetes into Class 1, while Class 0 represented no reported diabetes. Class 1 contained **35,346 records (13.93%)**, and Class 0 contained 218,334 records (86.07%).

Validation checks covered missing values, data types, target values, and feature ranges. No missing or out-of-range values were identified. Variables stored as floating-point values despite having integer-valued entries were converted to integer format.

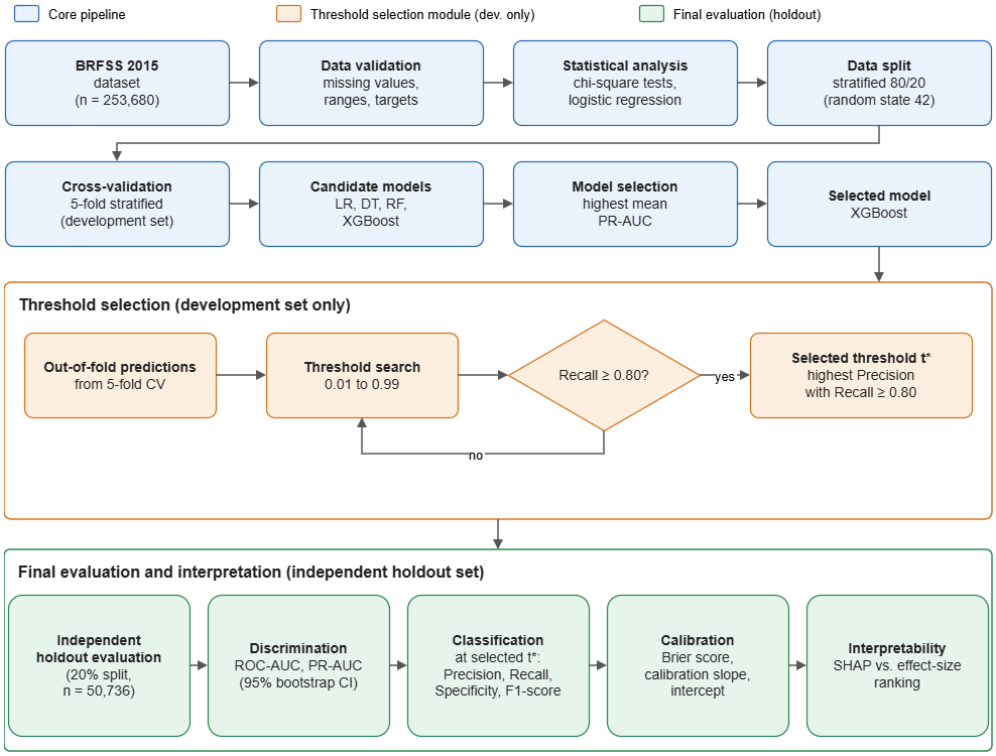



**Fig. 1** Analysis workflow used for statistical inference, model selection, holdout evaluation, calibration assessment, and model explanation.

A total of **24,206 repeated feature profiles** were identified. They were retained because the dataset contained no respondent identifier, so identical feature profiles could not be confirmed as duplicate individuals.

No SMOTE, oversampling, or undersampling was applied. All analyses therefore used the observed class distribution.

## 3.2 Statistical Association Analysis

Univariate tests were selected according to variable type. Eighteen binary, categorical, or ordinal predictors were evaluated using the Chi-Square Test of Independence. Association strength was measured with Cramér's $V$:

$$V = \sqrt{\frac{\chi^2}{N \cdot \min(r-1, c-1)}}. \tag{1}$$

For interpretation, $V < 0.05$ was treated as negligible, 0.05–$<$ 0.10 as weak, 0.10–$<$ 0.20 as small, 0.20–$<$ 0.30 as moderate, and $\geq 0.30$ as strong.

BMI, mental unhealthy days, and physical unhealthy days were treated as numerical variables. Group differences were evaluated using Welch's two-sample $t$-test and the Mann–Whitney U test. Cohen's $d$, rank-biserial correlation, and common-language effect size were used to describe effect magnitude. Absolute rank-biserial correlation was used when ranking numerical predictors by univariate effect size.

Holm correction was applied to the 18 Chi-Square tests and the three Mann–Whitney U tests. Effect sizes were interpreted alongside adjusted $p$-values because the large sample size could produce small $p$-values for weak associations.

A multivariable Logistic Regression model containing all 21 predictors was also fitted to estimate adjusted associations. Reported quantities included regression coefficients, adjusted odds ratios, 95% confidence intervals, Holm-adjusted $p$-values, and variance inflation factors. For binary predictors, an odds ratio compared values 1 and 0. For ordinal or numerical predictors, it represented a one-unit increase in the encoded

variable. These estimates were treated as sample-level associations and not as causal effects.

### 3.3 Model Development and Selection

The dataset was divided using a stratified 80/20 split with random state 42. The development set contained **202,944 records**, and the independent internal holdout set contained **50,736 records**. The holdout set was excluded from model selection, feature selection, threshold selection, and model-fitting decisions.

Four classifiers were compared using scikit-learn and XGBoost implementations [2, 17]: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Five-fold stratified cross-validation was performed only on the development set.

For predictive Logistic Regression, BMI, mental unhealthy days, and physical unhealthy days were standardized. GenHlth, Age, Education, and Income were one-hot encoded, while binary variables were retained without scaling. Tree-based models used the original encoded feature values.

The prespecified model configurations were: Logistic Regression with a maximum of 1,000 iterations; Decision Tree with maximum depth 8; Random Forest with 100 trees and maximum depth 12; and XGBoost with 100 boosting rounds, maximum depth 6, and learning rate 0.1.

Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC were calculated for each fold. **Mean PR-AUC was the primary model-selection metric** because Class 1 represented 13.93% of the dataset. Mean ROC-AUC and lower PR-AUC variability were prespecified as tie-breakers.

### 3.4 Threshold Selection and Holdout Evaluation

After model selection, out-of-fold probabilities were generated for every record in the development set. Candidate thresholds from 0.01 to 0.99 were evaluated without using the holdout set.

Thresholds satisfying

$$\text{Recall} \geq 0.80$$

were retained. Among these candidates, the threshold with the highest Precision was selected. F1-score and then the higher threshold were used as tie-breakers.

After the model and threshold were fixed, the selected classifier was trained on the complete development set and evaluated once on the independent holdout set. Reported metrics included Accuracy, Precision, Recall, Specificity, F1-score, ROC-AUC, PR-AUC, and the confusion matrix.

Uncertainty was estimated with **1,000 stratified bootstrap samples** drawn from the holdout set. Percentile-based 95% confidence intervals were calculated for ROC-AUC, PR-AUC, Recall, Precision, and F1-score.

### 3.5 Calibration and SHAP Analysis

Calibration was assessed on the holdout set using the Brier score, calibration slope, calibration intercept, and a reliability curve based on 10 probability groups. The slope and intercept were estimated from a logistic model relating the observed outcome to the log-odds of the predicted probabilities. No Platt scaling, isotonic regression, or other recalibration procedure was applied.

SHAP was used to explain the final fitted model. The original stratified split was reconstructed, and up to 10,000 holdout records were sampled using random state 42. For the selected tree-based model, TreeExplainer was used to calculate SHAP values. Global importance was defined as the mean absolute SHAP value for each predictor.

The SHAP ranking was compared with the univariate effect-size ranking. Cramér's $V$ was used for categorical and ordinal predictors, while absolute rank-biserial correlation was used for numerical predictors. A marginal association was treated as non-negligible when Cramér's $V \geq 0.05$ or $|r_{\text{rb}}| \geq 0.10$.

Agreement was evaluated using Top-5, Top-10, and Top-15 overlap, Jaccard similarity, and Spearman rank correlation. This comparison was used to contrast marginal statistical association with predictive contribution to the fitted model; it was not treated as causal or clinical validation of SHAP.

# 4 Results

## 4.1 Statistical Association Results

Among the categorical and ordinal variables, the largest associations with the target were observed for ***GenHlth*** ($V = 0.2993$), ***HighBP*** ($V = 0.2631$), *DiffWalk* ($V = 0.2183$), and *HighChol* ($V = 0.2003$). Age showed a smaller association ($V = 0.1862$). Several other variables remained statistically significant after Holm correction but had weak or negligible effect sizes.

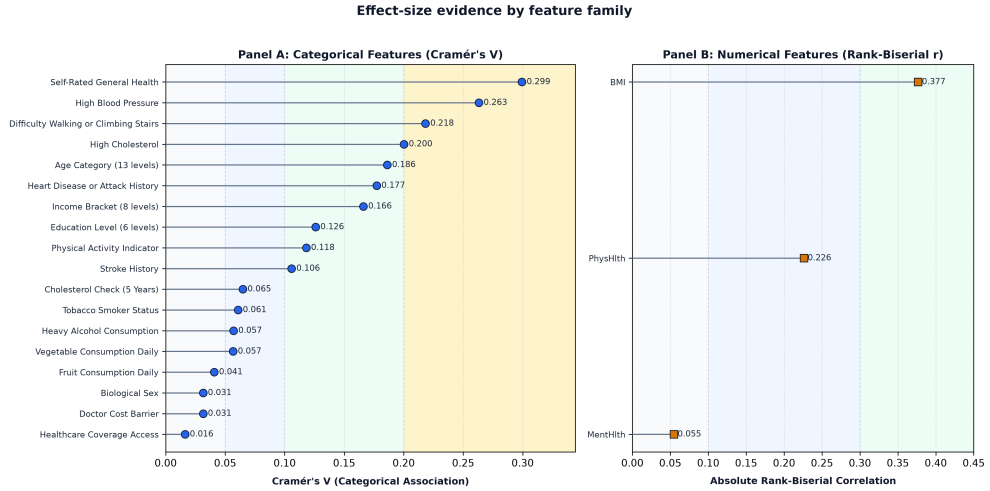



**Fig. 2** Univariate effect-size rankings by feature family. Categorical and ordinal predictors are ranked by Cramér's $V$, while numerical predictors are ranked by absolute rank-biserial correlation. Larger values indicate stronger marginal association within the analyzed sample.

For the numerical variables, BMI showed the largest difference between the two classes, followed by physical unhealthy days. Mental unhealthy days had a smaller distributional effect (Table 1).

**Table 1** Numerical Feature Comparisons Between Target Classes

| Variable | Class 0 | Class 1 | Cohen's $d$ | Rank-biserial $r$ | CLES |
|---|---|---|---|---|---|
| **BMI** | 27.81 | **31.94** | **0.6414** | **0.3766** | **0.6883** |
| PhysHlth | 3.64 | 7.95 | 0.5022 | 0.2260 | 0.6130 |
| MentHlth | 2.98 | 4.46 | 0.2006 | 0.0546 | 0.5273 |

*Note.* Class 0 denotes no reported diabetes; Class 1 denotes reported prediabetes or diabetes. All three Mann–Whitney U comparisons remained significant after Holm correction.

The multivariable Logistic Regression included all 21 predictors. Seventeen predictors remained statistically significant after Holm correction, and the maximum VIF was **1.7959**. The largest positive adjusted odds ratios among the main predictors were observed for ***CholCheck*, OR = 3.4695** (95% CI: 3.0331–3.9686), ***HighBP*, OR = 2.1331** (95% CI: 2.0723–2.1957), *HighChol*, OR = 1.7834, and *GenHlth*, OR = 1.7091.

For binary predictors, the odds ratio compares values 1 and 0. For ordinal or numerical predictors, it represents a one-unit increase in the encoded variable. These estimates describe adjusted associations within the analyzed sample and do not establish causal effects.

## 4.2 Cross-Validated Model Selection

The four classifiers were compared using five-fold stratified cross-validation on the development set, with mean PR-AUC as the primary selection metric.

XGBoost achieved the highest mean **PR-AUC of 0.4359** and **ROC-AUC of 0.8305**. Random Forest had the second-highest PR-AUC at 0.4301, a difference of 0.0058 from XGBoost.

**Table 2** Five-Fold Cross-Validation Performance on the Development Set

| Model | PR-AUC (mean ± SD) | ROC-AUC (mean ± SD) |
|---|---|---|
| Logistic Regression | 0.4163 ± 0.0052 | 0.8250 ± 0.0016 |
| Decision Tree | 0.4015 ± 0.0065 | 0.8169 ± 0.0025 |
| Random Forest | 0.4301 ± 0.0070 | 0.8271 ± 0.0024 |
| **XGBoost** | **0.4359 ± 0.0077** | **0.8305 ± 0.0021** |

*Note.* Values are mean ± standard deviation across five stratified folds. PR-AUC was the prespecified primary model-selection metric; mean ROC-AUC and then lower PR-AUC variability were used only as tie-breakers.

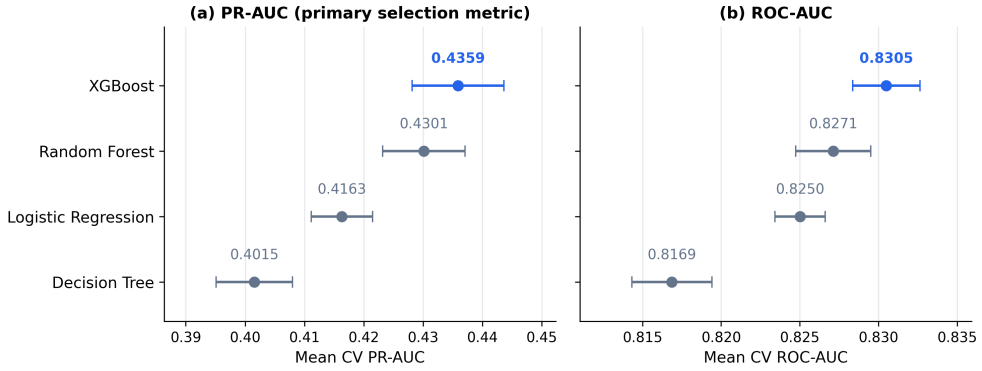



**Fig. 3** Cross-validated PR-AUC and ROC-AUC for the four classifiers.

XGBoost was therefore selected for threshold analysis and final holdout evaluation.

## 4.3 Threshold Selection, Holdout Performance, and Calibration

The classification threshold was selected from out-of-fold development predictions before the holdout set was evaluated. Among thresholds satisfying Recall $\geq 0.80$, the highest Precision was obtained at **0.13**.

On the independent internal holdout set, lowering the threshold from 0.50 to 0.13 increased Recall from 0.1654 to **0.8099**, while Precision decreased from 0.5583 to **0.2991**. False negatives decreased from 5,900 to **1,344**, whereas false positives increased from 925 to **13,416**.

**Table 3** Holdout Performance at the Default and Selected Thresholds

| Metric | Threshold 0.50 | **Threshold 0.13** |
|---|---|---|
| Precision | 0.5583 | 0.2991 |
| Recall | 0.1654 | **0.8099** |
| Specificity | 0.9788 | 0.6928 |
| F1-score | 0.2552 | **0.4369** |
| False negatives | 5,900 | **1,344** |
| False positives | 925 | 13,416 |

The threshold-independent holdout metrics were **ROC-AUC = 0.8272** and **PR-AUC = 0.4238**. Their 95% bootstrap confidence intervals were 0.8223–0.8322 and 0.4135–0.4347, respectively. At threshold 0.13, the bootstrap intervals were 0.8004–0.8194 for Recall, 0.2953–0.3028 for Precision, and 0.4317–0.4418 for F1-score.

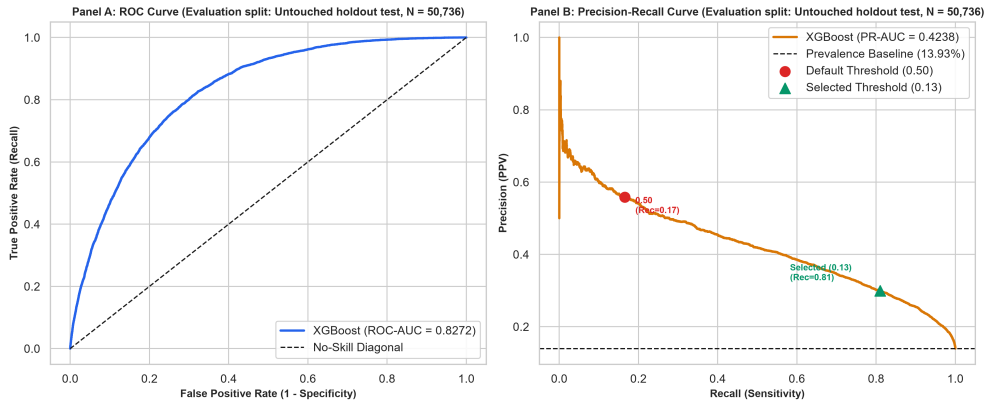



**Fig. 4** ROC and Precision–Recall curves for XGBoost on the independent internal holdout set.

Calibration on the holdout set yielded a **Brier score of 0.0974**, a **calibration slope of 0.9590**, and an **intercept of** −0.0514. No recalibration procedure was applied.

### 4.4 SHAP Explanation and Evidence Alignment

The five highest-ranked features by mean absolute SHAP value were ***GenHlth***, ***HighBP***, **Age**, **BMI**, and ***HighChol***. Their mean absolute SHAP values were 0.6381, 0.5264, 0.3994, 0.3978, and 0.2927, respectively.

Higher encoded *GenHlth* scores, high blood pressure, older age, and higher BMI more often produced positive SHAP contributions. These values describe contributions to the fitted model prediction and do not estimate causal effects.

Nine of the Top-10 SHAP features met the prespecified threshold for a non-negligible univariate association. However, only **7 of the Top-10 features** were shared between the SHAP and univariate effect-size rankings, corresponding to a **Jaccard similarity of 0.5385**.

Across all 21 predictors, the two rankings had a **Spearman correlation of** $r_s = 0.7098$ ($p = 3.13 \times 10^{-4}$). The Top-5, Top-10, and Top-15 overlaps were 4/5, 7/10, and 13/15, respectively.

Differences between the two rankings were observed for several predictors. For example, *DiffWalk* had Cramér's $V = 0.2183$ but ranked 12th by SHAP, whereas Sex ranked seventh by SHAP despite a marginal Cramér's $V = 0.0314$. The two rankings therefore quantify different properties: marginal association and contribution to the fitted multivariable model.

## 5 Discussion

Three findings require interpretation. First, XGBoost was selected under the prespecified cross-validation rule, although its mean PR-AUC differed from Random Forest by only 0.0058. Second, the development-selected threshold of 0.13 substantially changed the balance between missed positive cases and false-positive predictions. Third, SHAP and univariate effect-size rankings agreed on several major predictors but did not produce identical rankings.

### 5.1 Model Selection, Screening Threshold, and Calibration

XGBoost had the highest mean cross-validated PR-AUC, but the absolute difference from Random Forest was small. The selection should therefore be interpreted as

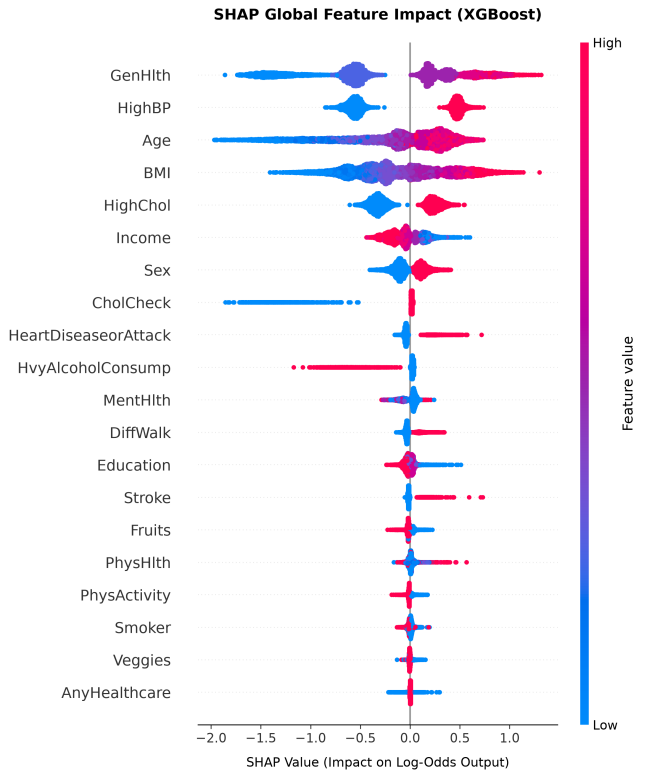



**Fig. 5** SHAP summary plot for the final XGBoost model.

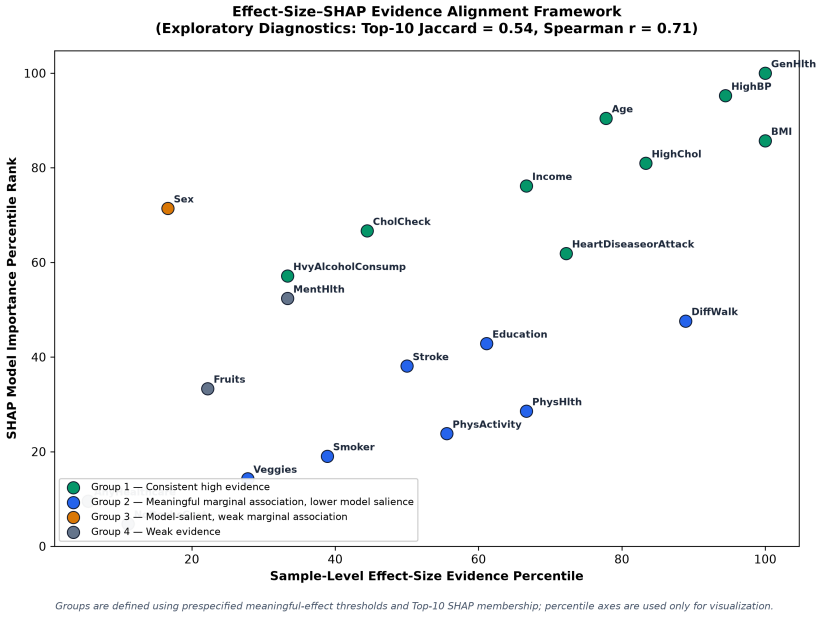



**Fig. 6** Comparison between univariate effect-size ranking and global SHAP importance.

the outcome of the prespecified model-selection rule rather than evidence of a large performance advantage over Random Forest.

The threshold analysis produced a larger practical change than the choice between the two highest-ranked models. At the default threshold of 0.50, the model detected only 16.54% of the positive holdout records. Using the development-selected threshold of **0.13 increased Recall to 80.99%**, while **Precision decreased to 29.91%**. False negatives decreased from 5,900 to 1,344, whereas false positives increased from 925 to 13,416.

This threshold therefore represents a screening-oriented operating point under the study's prespecified Recall constraint. It should not be interpreted as a general clinical cutoff. A setting in which false-positive follow-up is costly may require a different threshold.

The threshold was selected exclusively from out-of-fold development predictions before holdout evaluation. This separation prevents direct optimization of the decision threshold on the reported holdout results.

Calibration estimates on the internal holdout were a Brier score of 0.0974, a slope of **0.9590**, and an intercept of $-0.0514$. The slope and intercept were close to their reference values of 1 and 0, respectively, within this holdout sample. These estimates do not establish calibration in another year, population, or clinical setting. No recalibration procedure was evaluated in the present analysis.

## 5.2 Statistical Associations and SHAP Explanations

The statistical and SHAP analyses identified several of the same predictors near the top of their rankings. *GenHlth*, *HighBP*, Age, BMI, and *HighChol* were among the highest-ranked predictors in the final XGBoost model and also showed non-negligible univariate associations with the target.

However, the two analyses measure different quantities. Univariate effect sizes describe marginal differences between target classes, whereas SHAP values describe contributions to predictions made by the fitted multivariable model. Accordingly, exact agreement between the rankings is not expected.

The Top-10 rankings shared **7 features**, with a **Jaccard similarity of 0.5385**. Across all 21 predictors, the Spearman rank correlation was $r_s = 0.7098$.

Several predictors illustrate the difference between the two ranking systems. *DiffWalk* had a Cramér's $V$ of 0.2183 but ranked 12th by SHAP, and *PhysHlth* had a rank-biserial correlation of 0.2260 but ranked 16th by SHAP. Conversely, Sex ranked seventh by SHAP despite a marginal Cramér's $V$ of 0.0314.

These differences can arise when predictors contain overlapping information or participate in nonlinear model structures, but the present analysis does not identify the specific mechanism responsible for each ranking difference. SHAP values should therefore be interpreted as explanations of model behavior rather than estimates of biological importance or causal effects.

## 5.3 Relation to Previous BRFSS Studies

The proposed framework emphasizes screening sensitivity rather than maximizing overall accuracy. On the same independent holdout set, selecting the threshold from development-set out-of-fold predictions increased Recall from 16.54% to **80.99%**, reducing missed positive cases by **77.22%**. This improvement came at the cost of lower Precision (29.91%) and more false-positive predictions. Although several previous BRFSS studies reported higher classification performance under different experimental conditions, the main contribution of this work lies in its leakage-controlled threshold selection, explicit screening trade-off, and reliability-oriented evaluation rather than superiority in cross-study predictive accuracy.

The main contribution of the present study is consequently the evaluation sequence rather than a new classifier: cross-validated model selection on the development set, threshold selection from out-of-fold development predictions, one-time internal

**Table 4** Screening-oriented comparison with selected BRFSS studies. Cross-study results are descriptive due to differences in experimental settings.

| Study | Dataset / Model | AUC (%) | Recall (%) |
|---|---|---|---|
| *Previously published studies* | | | |
| Xie et al. [23] | BRFSS 2014; Neural Network | 79.49 | 37.81 |
| Mohamed et al. [12] | BRFSS 2015; XGBoost | 82.00 | 19.00 |
| Li et al. [7] | BRFSS-derived; Unbalanced XGBoost | 82.82 | 17.34 |
| Muhammad et al. [15] | Tennessee 2023; Gradient Boosting | 80.00 | 32.00 |
| Majcherek et al. [11] | BRFSS 2015; Extra Trees / Resampling | Up to 99 | 99.00 |
| **Present study: same independent holdout** | | | |
| Present study | XGBoost; Default ($t = 0.50$) | 82.72 | 16.54 |
| **Present study** | **XGBoost; Selected ($t = 0.13$)** | **82.72** | **80.99** |

**Within-study screening improvement:** Recall increased by **64.45 percentage points**, while missed positive cases decreased by **77.22%** (5,900 to 1,344).

*Trade-off:* Precision decreased from 55.83% to 29.91%.

*Note.* Published results use different datasets, class-balancing strategies, and evaluation protocols and are not directly rankable. Li et al.'s results shown here refer to the unbalanced XGBoost setting; Majcherek et al.'s high reported results involve different resampling and evaluation configurations. The threshold comparison for the present study uses the same holdout predictions.

holdout evaluation, bootstrap uncertainty, calibration assessment, and an explicitly exploratory comparison between SHAP rankings and univariate effect sizes. This framing is consistent with current guidance that emphasizes transparent model evaluation and reporting rather than performance claims based on a single metric [4, 14].

### 5.4 Limitations and Future Work

Several limitations affect the interpretation of the results. BRFSS is cross-sectional and self-reported, so the observed associations cannot establish disease causation or future diabetes onset. The positive class combines reported prediabetes and diabetes, while Class 0 indicates no reported diabetes rather than laboratory-confirmed absence of disease.

The analyzed dataset also lacks survey weights and design variables, respondent identifiers, and clinical biomarkers such as HbA1c or fasting glucose. The reported estimates therefore describe the analyzed unweighted sample and should not be interpreted as population-level estimates for the United States.

Evaluation was limited to an internal holdout from the same 2015 dataset. The four predictive models used fixed configurations rather than nested hyperparameter optimization. In addition, repeated feature profiles were retained because respondent identifiers were unavailable. Identical profiles could therefore occur in both development and holdout sets even when they represent different respondents.

The threshold of 0.13 was selected for the specific constraint Recall $\geq 0.80$. Its suitability depends on the relative cost of missed positive cases and false-positive follow-up. The SHAP alignment analysis also depends on the selected effect-size definitions and Top-$k$ cutoffs.

Future evaluation should prioritize temporal or external validation, including newer BRFSS waves or datasets with clinical biomarkers. Additional analyses should examine subgroup performance, SHAP stability, the effect of repeated profiles on validation estimates, survey-weighted inference, and nested model tuning.

## 6 Conclusion

A reliability-oriented and interpretable diabetes analytics workflow was developed using the original imbalanced CDC BRFSS 2015 dataset with 253,680 records.

Repeated response profiles were retained because respondent identifiers were unavailable and identical records could not be confirmed as duplicate individuals. Among four evaluated models, XGBoost achieved the highest mean five-fold cross-validated PR-AUC of 0.4359 and was selected for final evaluation.

On the independent internal holdout set, XGBoost achieved a PR-AUC of 0.4238 and a ROC-AUC of 0.8272, with bootstrap 95% confidence intervals of 0.4135–0.4347 and 0.8223–0.8322, respectively. The screening threshold of 0.13 was selected from out-of-fold development predictions before holdout evaluation. At this threshold, Recall reached 80.99% (95% CI: 80.04%–81.94%), Precision reached 29.91% (95% CI: 29.53%–30.28%), Specificity was 69.28%, and F1-score reached 0.4369 (95% CI: 0.4317–0.4418). False negatives decreased from 5,900 to 1,344, although lower Precision and Specificity produced more false-positive referrals. Calibration was reasonably stable, with a Brier score of 0.0974, a calibration slope of 0.9590, and an intercept of $-0.0514$.

Adjusted Logistic Regression results supported several sample-level associations after accounting for all predictors, while variance inflation factors remained below 1.80, indicating no severe multicollinearity. SHAP identified self-rated general health, high blood pressure, age, BMI, and high cholesterol as the most influential predictive features. Seven of the Top-10 SHAP and statistical effect-size features overlapped, producing a Jaccard similarity of 0.5385 and a Spearman correlation of 0.7098. This agreement was meaningful but incomplete, reflecting differences between marginal statistical association and multivariable predictive contribution. Overall, the workflow combines statistical analysis, leakage-controlled model and threshold selection, independent holdout evaluation, uncertainty estimation, calibration, and model explanation. The results support screening-oriented analysis but should not be interpreted as clinical diagnosis, causal evidence, or national population estimates. External validation on newer BRFSS waves, other populations, and datasets containing clinical biomarkers remains necessary.

## Declaration

Declaration of generative AI and AI-assisted technologies in the writing process: generative AI tools were used for language editing only. The authors reviewed and revised the manuscript and take full responsibility for the accuracy and integrity of the work.

## References


[1] Centers for Disease Control and Prevention (CDC) (2015) Behavioral risk factor surveillance system (brfss). CDC Public Health Data

[2] Chen T, Guestrin C (2016) XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp 785–794

[3] Collins GS, Dhiman P, Ma J, et al (2024) Evaluation of clinical prediction models (part 1): From development to external validation. The BMJ 384:e074819

[4] Collins GS, Moons KGM, Dhiman P, et al (2024) TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. The BMJ 385:e078378

[5] Efthimiou O, Seo M, Chalkou K, et al (2024) Developing clinical prediction models: A step-by-step guide. The BMJ 386:e078276

[6] Huang Y, Li W, Macheret F, et al (2020) A tutorial on calibration measurements and calibration models for clinical prediction models. Journal of the American Medical Informatics Association 27(4):621–633

[7] Li W, Peng Y, Peng K (2024) Diabetes prediction model based on GA-XGBoost and stacking ensemble algorithm. PLOS ONE 19(9):e0311222

[8] Lugner M, Rawshani A, Helleryd E, et al (2024) Identifying top ten predictors of type 2 diabetes through machine learning analysis of UK Biobank data. Scientific Reports 14:2102

[9] Lundberg SM, Lee SI (2017) A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems

[10] Lundberg SM, Erion G, Chen H, et al (2020) From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence 2(1):56–67

[11] Majcherek D, Ciesielski A, Sobczak P (2025) AI-driven analysis of diabetes risk determinants in U.S. adults: Exploring disease prevalence and health factors. PLOS ONE 20(9):e0328655

[12] Mohamed MH, Khafagy MH, Kamel NMM, et al (2024) Diabetic mellitus prediction with brfss data sets. Journal of Theoretical and Applied Information Technology 102(3):883–897

[13] Mohsen F, Al-Absi HRH, Yousri NA, et al (2023) A scoping review of artificial intelligence-based methods for diabetes risk prediction. npj Digital Medicine 6:197

[14] Moons KGM, Damen JAA, Kaul T, et al (2025) PROBAST+AI: An updated quality, risk of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods. The BMJ 388:e082505

[15] Muhammad MA, Sani J, Ahmed MM (2025) Exploring explainable machine learning for predicting and interpreting self-reported diabetes among tennessee adults: Insights from the 2023 behavioral risk factor surveillance system (BRFSS). Journal of Primary Care & Community Health 16:21501319251400546

[16] Olisah CC, Smith L, Smith M (2022) Diabetes mellitus prediction and diagnosis from a data preprocessing and machine learning perspective. Computer Methods and Programs in Biomedicine 220:106773

[17] Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: Machine learning in python. Journal of Machine Learning Research 12(Oct):2825–2830

[18] Prendin F, Pavan J, Cappon G, et al (2023) The importance of interpreting machine learning models for blood glucose prediction in diabetes: An analysis using SHAP. Scientific Reports 13:16865

[19] Saito T, Rehmsmeier M (2015) The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLOS ONE 10(3):e0118432

[20] Teboul A (2021) Diabetes health indicators dataset. Kaggle

[21] UCI Machine Learning Repository (2023) Cdc diabetes health indicators dataset

[22] Ullah Z, Saleem F, Jamjoom M, et al (2022) Detecting high-risk factors and early diagnosis of diabetes using machine learning methods. Computational Intelligence and Neuroscience 2022:2557795

[23] Xie Z, Nikolayeva O, Luo J, et al (2019) Building risk prediction models for type 2 diabetes using machine learning techniques. Preventing Chronic Disease 16:E130

[24] Zhang X, Yao W, Wang D, et al (2024) Development and validation of machine learning models for identifying prediabetes and diabetes in normoglycemia. Diabetes/Metabolism Research and Reviews 40(8):e70003